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MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE

$$M_{2\times 2} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$D_{2\times 2} = ad - bc$$

Study Material

Determinant

(English Medium)

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DETERMINANT

1. INTRODUCTION :

If the equations $a_1x + b_1 = 0$, $a_2x + b_2 = 0$ are satisfied by the same value of x, then $a_1b_2 - a_2b_1 = 0$. The expression $a_1b_2 - a_2b_1$ is called a determinant of the second order, and is denoted by :

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

A determinant of second order consists of two rows and two columns.

Next consider the system of equations $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$, $a_3x + b_3y + c_3 = 0$

If these equations are satisfied by the same values of x and y, then on eliminating x and y we get.

$$a_1(b_2c_3 - b_3c_2) + b_1(c_2a_3 - c_3a_2) + c_1(a_2b_3 - a_3b_2) = 0$$

The expression on the left is called a determinant of the third order, and is denoted by

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

A determinant of third order consists of three rows and three columns.

2. VALUE OF A DETERMINANT :

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1\begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1\begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

Note : Sarrus diagram to get the value of determinant of order three :

$$\begin{array}{c} \quad -ve\ -ve\ -ve \\ D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \left|\begin{matrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{matrix}\right| \begin{matrix} a_1 & b_1 \\ a_2 & b_2 \\ a_3 & b_3 \end{matrix} = (a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2) - (a_3b_2c_1 + a_2b_1c_3 + a_1b_3c_2) \\ \quad +ve\ +ve\ +ve \end{array}$$

Note that the product of the terms in first bracket (i.e. $a_1a_2a_3b_1b_2b_3c_1c_2c_3$) is same as the product of the terms in second bracket.

Illustration 1 : The value of $\begin{vmatrix} 1 & 2 & 3 \\ -4 & 3 & 6 \\ 2 & -7 & 9 \end{vmatrix}$ is -

(A) 213 (B) – 231 (C) 231 (D) 39

Solution :

$$\begin{vmatrix} 1 & 2 & 3 \\ -4 & 3 & 6 \\ 2 & -7 & 9 \end{vmatrix} = 1\begin{vmatrix} 3 & 6 \\ -7 & 9 \end{vmatrix} - 2\begin{vmatrix} -4 & 6 \\ 2 & 9 \end{vmatrix} + 3\begin{vmatrix} -4 & 3 \\ 2 & -7 \end{vmatrix}$$

$= (27 + 42) - 2(-36 - 12) + 3(28 - 6) = 231$

Alternative : By sarrus diagram

$$\begin{vmatrix} 1 & 2 & 3 \\ -4 & 3 & 6 \\ 2 & -7 & 9 \end{vmatrix} = \left|\begin{matrix} 1 & 2 & 3 \\ -4 & 3 & 6 \\ 2 & -7 & 9 \end{matrix}\right| \begin{matrix} 1 & 2 \\ -4 & 3 \\ 2 & -7 \end{matrix}$$

$= (27 + 24 + 84) - (18 - 42 - 72) = 135 - (18 - 114) = 231$ **Ans. (C)**

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3. MINORS & COFACTORS :

The minor of a given element of determinant is the determinant obtained by deleting the row & the column in which the given element stands.

For example, the minor of a_1 in $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ is $\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}$ & the minor of b_2 is $\begin{vmatrix} a_1 & c_1 \\ a_3 & c_3 \end{vmatrix}$.

Hence a determinant of order three will have "9 minors".

If M_{ij} represents the minor of the element belonging to i^{th} row and j^{th} column then the cofactor of that element is given by : $C_{ij} = (-1)^{i+j}. M_{ij}$

Illustration 2 : Find the minors and cofactors of elements '–3', '5', '–1' & '7' in the determinant $\begin{vmatrix} 2 & -3 & 1 \\ 4 & 0 & 5 \\ -1 & 6 & 7 \end{vmatrix}$

Solution :

Minor of $-3 = \begin{vmatrix} 4 & 5 \\ -1 & 7 \end{vmatrix} = 33$; Cofactor of $-3 = -33$

Minor of $5 = \begin{vmatrix} 2 & -3 \\ -1 & 6 \end{vmatrix} = 9$; Cofactor of $5 = -9$

Minor of $-1 = \begin{vmatrix} -3 & 1 \\ 0 & 5 \end{vmatrix} = -15$; Cofactor of $-1 = -15$

Minor of $7 = \begin{vmatrix} 2 & -3 \\ 4 & 0 \end{vmatrix} = 12$; Cofactor of $7 = 12$

4. EXPANSION OF A DETERMINANT IN TERMS OF THE ELEMENTS OF ANY ROW OR COLUMN:

Let $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

(i) The sum of the product of elements of any row (column) with their corresponding cofactors is always equal to the value of the determinant.

D can be expressed in any of the six forms :

$a_1A_1 + b_1B_1 + c_1C_1$, $\quad a_1A_1 + a_2A_2 + a_3A_3$,

$a_2A_2 + b_2B_2 + c_2C_2$, $\quad b_1B_1 + b_2B_2 + b_3B_3$,

$a_3A_3 + b_3B_3 + c_3C_3$, $\quad c_1C_1 + c_2C_2 + c_3C_3$,

where A_i, B_i & C_i $(i = 1,2,3)$ denote cofactors of a_i, b_i & c_i respectively.

(ii) The sum of the product of elements of any row (column) with the cofactors of other row (column) is always equal to zero.

Hence,

$a_2A_1 + b_2B_1 + c_2C_1 = 0$,

$b_1A_1 + b_2A_2 + b_3A_3 = 0$ and so on.

where A_i, B_i & C_i $(i = 1,2,3)$ denote cofactors of a_i, b_i & c_i respectively.

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Do yourself -1 :

1. Find minors & cofactors of elements '6', '5', '0' & '4' of the determinant $\begin{vmatrix} 2 & 1 & 3 \\ 6 & 5 & 7 \\ 3 & 0 & 4 \end{vmatrix}$.

2. Calculate the value of the determinant $\begin{vmatrix} 5 & -3 & 7 \\ -2 & 4 & -8 \\ 9 & 3 & -10 \end{vmatrix}$

3. The value of the determinant $\begin{vmatrix} a & b & 0 \\ 0 & a & b \\ b & 0 & a \end{vmatrix}$ is equal to -

(A) $a^3 - b^3$ (B) $a^3 + b^3$ (C) 0 (D) none of these

4. Find the value of 'k', if $\begin{vmatrix} 1 & 2 & 0 \\ 2 & 3 & 1 \\ 3 & k & 2 \end{vmatrix} = 4$

5. Which of the following is correct?

(A) $\begin{vmatrix} 2 & 4 \\ -5 & -1 \end{vmatrix} = 17$ (B) $\begin{vmatrix} 3 & -1 & -2 \\ 0 & 0 & -1 \\ 3 & -5 & 0 \end{vmatrix} = 12$

(C) $\begin{vmatrix} 3 & -4 & 5 \\ 1 & 1 & -2 \\ 2 & 3 & 1 \end{vmatrix} = 46$ (D) $\begin{vmatrix} 2 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{vmatrix} = 4$

Comprehension (For Q.6 to 8)

Consider the determinant, $\Delta = \begin{vmatrix} p & q & r \\ x & y & z \\ \ell & m & n \end{vmatrix}$

M_{ij} denotes the minor of an element in i^{th} row and j^{th} column

C_{ij} denotes the cofactor of an element in i^{th} row and j^{th} column

6. The value of $p.C_{21} + q.C_{22} + r.C_{23}$ is

(A) 0 (B) $-\Delta$ (C) Δ (D) None of these

7. The value of $x.C_{21} + y.C_{22} + z.C_{23}$ is

(A) 0 (B) $-\Delta$ (C) Δ (D) None of these

8. The value of $q.M_{12} - y.M_{22} + m.M_{32}$ is

(A) 0 (B) $-\Delta$

(C) $x.M_{21} - y.M_{22} + z.M_{23}$ (D) Both (B) & (C)

9. If the minor of three-one element (i.e. M_{31}) in the determinant $\begin{vmatrix} 0 & 1 & \sec\alpha \\ \tan\alpha & -\sec\alpha & \tan\alpha \\ 1 & 0 & 1 \end{vmatrix}$ is 1

then find the value of $\alpha.(0 \le \alpha \le \pi)$

(A) 0 (B) $\frac{3\pi}{4}$ (C) π (D) All of these

10. If $f(x) = \begin{vmatrix} x-1 & x+1 \\ x-2 & x+2 \end{vmatrix}$, then find the value of $\begin{vmatrix} f(-1) & f(-2) \\ f(1) & f(2) \end{vmatrix}$.

(A) 0 (B) 1 (C) 3 (D) 4

5. PROPERTIES OF DETERMINANTS :

(a) The value of a determinant remains unaltered, if the rows & columns are inter-changed,

e.g. if $D=\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}=\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

(b) If any two rows (or columns) of a determinant be interchanged, the value of determinant is changed in sign only. e.g.

Let $D=\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ & $D_1=\begin{vmatrix} a_2 & b_2 & c_2 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix}$. Then $D_1=-D$.

(c) If all the elements of a row (or column) are zero, then the value of the determinant is zero.

(d) If all the elements of any row (or column) are multiplied by the same number, then the determinant is multiplied by that number.

e.g. If $D=\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $D_1=\begin{vmatrix} Ka_1 & Kb_1 & Kc_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$. Then $D_1 = KD$

(e) If all the elements of a row (or column) are proportional (or identical) to the element of any other row, then the determinant vanishes, i.e. its value is zero.

e.g. If $D=\begin{vmatrix} a_1 & b_1 & c_1 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix}\Rightarrow D=0$; If $D_1=\begin{vmatrix} a_1 & b_1 & c_1 \\ ka_1 & kb_1 & kc_1 \\ a_3 & b_3 & c_3 \end{vmatrix}\Rightarrow D_1=0$

Illustration 3 : Prove that $\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix}=\begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$

Solution : $D=\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix}=\begin{vmatrix} a & x & p \\ b & y & q \\ c & z & r \end{vmatrix}$ (By interchanging rows & columns)

$=-\begin{vmatrix} x & a & p \\ y & b & q \\ z & c & r \end{vmatrix}$ $(C_1 \leftrightarrow C_2)$

$=\begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$ $(R_1 \leftrightarrow R_2)$

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Illustration 4 : Find the value of the determinant $\begin{vmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{vmatrix}$

Solution : $D = \begin{vmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{vmatrix} = a\begin{vmatrix} a & b & c \\ ab & b^2 & bc \\ ac & bc & c^2 \end{vmatrix} = abc\begin{vmatrix} a & b & c \\ a & b & c \\ a & b & c \end{vmatrix} = 0$

Since all rows are same, hence value of the determinant is zero.

Do yourself -2 :

1. Without expanding the determinant prove that $\begin{vmatrix} a & p & \ell \\ b & q & m \\ c & r & n \end{vmatrix} + \begin{vmatrix} r & n & c \\ q & m & b \\ p & \ell & a \end{vmatrix} = 0$

2. If $D = \begin{vmatrix} \alpha & \beta \\ \gamma & \delta \end{vmatrix}$, then $\begin{vmatrix} 2\alpha & 2\beta \\ 2\gamma & 2\delta \end{vmatrix}$ is equal to -

(A) D (B) 2D (C) 4D (D) 16D

3. Find the value of $\begin{vmatrix} 53 & 106 & 159 \\ 52 & 65 & 91 \\ 102 & 153 & 221 \end{vmatrix}$.

4. Find the value of $\begin{vmatrix} 13 & 3 & 23 \\ 30 & 7 & 53 \\ 39 & 9 & 70 \end{vmatrix}$

(A) 1 (B) 3 (C) 5 (D) 8

5. The determinant $\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix}$ is same as:

(A) $\begin{vmatrix} a & b & p \\ y & b & q \\ z & c & r \end{vmatrix}$ (B) $\begin{vmatrix} x & b & q \\ y & a & p \\ z & c & r \end{vmatrix}$ (C) $\begin{vmatrix} x & z & y \\ p & r & q \\ a & c & b \end{vmatrix}$ (D) $\begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$

(f) If each element of any row (or column) is expressed as a sum of two (or more) terms, then the determinant can be expressed as the sum of two (or more) determinants.

e.g. $\begin{vmatrix} a_1+x & b_1+y & c_1+z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} x & y & z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

Note that : If $D_r = \begin{vmatrix} f(r) & g(r) & h(r) \\ a & b & c \\ a_1 & b_1 & c_1 \end{vmatrix}$

where $r \in N$ and a, b, c, a_1, b_1, c_1 are constants, then

$$\sum_{r=1}^{n} D_r = \begin{vmatrix} \sum_{r=1}^{n} f(r) & \sum_{r=1}^{n} g(r) & \sum_{r=1}^{n} h(r) \\ a & b & c \\ a_1 & b_1 & c_1 \end{vmatrix}$$

(g) **Row - column operation :** The value of a determinant remains unaltered under a column (C_i) operation of the form $C_i \to C_i + \alpha C_j + \beta C_k$ $(j, k \neq i)$ or row (R_i) operation of the form $R_i \to R_i + \alpha R_j + \beta R_k$ $(j, k \neq i)$. In other words, the value of a determinant is not altered by adding the elements of any row (or column) to the same multiples of the corresponding elements of any other row (or column)

e.g. Let $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

$$D = \begin{vmatrix} a_1+\alpha a_2 & b_1+\alpha b_2 & c_1+\alpha c_2 \\ a_2 & b_2 & c_2 \\ a_3+\beta a_2 & b_3+\beta b_2 & c_3+\beta c_2 \end{vmatrix} \quad (R_1 \to R_1 + \alpha R_2;\ R_3 \to R_3 + \beta R_2)$$

Note :

(i) By using the operation $R_i \to xR_i + yR_j + zR_k$ $(j, k \neq i)$, the value of the determinant becomes x times the original one.

(ii) While applying this property **ATLEAST ONE ROW (OR COLUMN)** must remain unchanged.

Illustration 5 : If $D_r = \begin{vmatrix} r & r^3 & 2 \\ n & n^3 & 2n \\ \frac{n(n+1)}{2} & \left(\frac{n(n+1)}{2}\right)^2 & 2(n+1) \end{vmatrix}$, find $\sum_{r=0}^{n} D_r$.

Solution :

$$\sum_{r=0}^{n} D_r = \begin{vmatrix} \sum_{r=0}^{n} r & \sum_{r=0}^{n} r^3 & \sum_{r=0}^{n} 2 \\ n & n^3 & 2n \\ \frac{n(n+1)}{2} & \left(\frac{n(n+1)}{2}\right)^2 & 2(n+1) \end{vmatrix} = \begin{vmatrix} \frac{n(n+1)}{2} & \left(\frac{n(n+1)}{2}\right)^2 & 2(n+1) \\ n & n^3 & 2n \\ \frac{n(n+1)}{2} & \left(\frac{n(n+1)}{2}\right)^2 & 2(n+1) \end{vmatrix} = 0$$ **Ans.**

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Illustration 6 : If $\begin{vmatrix} 3^2+k & 4^2 & 3^2+3+k \\ 4^2+k & 5^2 & 4^2+4+k \\ 5^2+k & 6^2 & 5^2+5+k \end{vmatrix} = 0$, then the value of k is-

(A) 2 (B) 1 (C) –1 (D) 0

Solution : Applying $(C_3 \to C_3 - C_1)$

$$D = \begin{vmatrix} 3^2+k & 4^2 & 3 \\ 4^2+k & 5^2 & 4 \\ 5^2+k & 6^2 & 5 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 9+k & 16 & 3 \\ 7 & 9 & 1 \\ 9 & 11 & 1 \end{vmatrix} = 0 \qquad (R_3 \to R_3 - R_2;\ R_2 \to R_2 - R_1)$$

$\Rightarrow \quad k - 1 = 0 \Rightarrow k = 1$ **Ans. (B)**

Do yourself - 3 :

1. Solve for x : $\begin{vmatrix} x & 2 & 0 \\ 2+x & 5 & -1 \\ 5-x & 1 & 2 \end{vmatrix} = 0$

2. If $D_r = \begin{vmatrix} 2r & 1 & n \\ 1 & -2 & 3 \\ 3 & 2 & 1 \end{vmatrix}$, then find the value of $\sum_{r=1}^{n} D_r$.

3. If $A + B + C = \pi$, then the value of the determinant $D = \begin{vmatrix} \sin^2 A & \cot A & 1 \\ \sin^2 B & \cot B & 1 \\ \sin^2 C & \cot C & 1 \end{vmatrix}$ is equal to

(A) 1 (B) –1 (C) 0 (D) none of these

4. The value of the determinant $\begin{vmatrix} \sin^2 13° & \sin^2 77° & \tan 135° \\ \sin^2 77° & \tan 135° & \sin^2 13° \\ \tan 135° & \sin^2 13° & \sin^2 77° \end{vmatrix}$ is equal to

(A) –1 (B) 0 (C) 3 (D) 7

5. If $D_r = \begin{vmatrix} 2^{r-1} & 2\left(3^{r-1}\right) & 4\left(5^{r-1}\right) \\ x & y & z \\ 2^n - 1 & 3^n - 1 & 5^n - 1 \end{vmatrix}$ then the value of $\sum_{r=1}^{n} D_r$ is

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(h) **Factor theorem :** If the elements of a determinant D are rational integral functions of x and two rows (or columns) become identical when x = a then (x – a) is a factor of D.

Note that if r rows become identical when a is substituted for x, then $(x-a)^{r-1}$ is a factor of D.

Illustration 7 : Prove that $\begin{vmatrix} a & a & x \\ m & m & m \\ b & x & b \end{vmatrix} = m(x-a)(x-b)$

Solution : Using factor theorem,

Put x = a

$$D = \begin{vmatrix} a & a & a \\ m & m & m \\ b & a & b \end{vmatrix} = 0$$

Since R_1 and R_2 are proportional which makes D = 0, therefore (x – a) is a factor of D.

Similarly, by putting x = b, D becomes zero, therefore (x – b) is a factor of D.

$$D = \begin{vmatrix} a & a & x \\ m & m & m \\ b & x & b \end{vmatrix} = \lambda(x-a)(x-b) \qquad ....(i)$$

To get the value of λ, put x = 0 in equation (i)

$$\begin{vmatrix} a & a & 0 \\ m & m & m \\ b & 0 & b \end{vmatrix} = \lambda ab$$

$amb = \lambda ab \quad \Rightarrow \quad \lambda = m$

$\therefore \quad D = m(x-a)(x-b)$

Do yourself - 4 :

1. Without expanding the determinant prove that $\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = (a-b)(b-c)(c-a)$

2. Using factor theorem, find the solution set of the equation $\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix} = 0$

3. $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix}$ is equals to

(A) (a – b) (b – c) (c – a) (ab + bc + ca)
(B) (a – b) (b – c) (c – a)
(C) (ab + bc + ca)
(D) None of these

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4. $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix}$ is equals to

(A) (a – b) (b – c) (c – a) (ab + bc + ca)
(B) (a – b) (b – c) (c – a)
(C) (a – b) (b – c) (c – a) (a + b + c)
(D) None of these

5. $\begin{vmatrix} a & b+c & a^2 \\ b & c+a & b^2 \\ c & a+b & c^2 \end{vmatrix}$ is equals to

(A) (a – b) (b – c) (c – a)
(B) –(a + b + c) (a – b) (b – c) (c – a)
(C) (a – b) (b – c) (c – a) (a + b + c)
(D) None of these

6. MULTIPLICATION OF TWO DETERMINANTS :

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \times \begin{vmatrix} l_1 & m_1 \\ l_2 & m_2 \end{vmatrix} = \begin{vmatrix} a_1 l_1 + b_1 l_2 & a_1 m_1 + b_1 m_2 \\ a_2 l_1 + b_2 l_2 & a_2 m_1 + b_2 m_2 \end{vmatrix}$$

Similarly two determinants of order three are multiplied.

(a) Here we have multiplied row by column. We can also multiply row by row, column by row and column by column.

(b) If D_1 is the determinant formed by replacing the elements of determinant D of order n by their corresponding cofactors then $D_1 = D^{n-1}$

Illustration 8 : Let α & β be the roots of equation $ax^2 + bx + c = 0$ and $S_n = \alpha^n + \beta^n$ for $n \ge 1$. Evaluate the value of the determinant $\begin{vmatrix} 3 & 1+S_1 & 1+S_2 \\ 1+S_1 & 1+S_2 & 1+S_3 \\ 1+S_2 & 1+S_3 & 1+S_4 \end{vmatrix}$.

Solution :

$$D = \begin{vmatrix} 3 & 1+S_1 & 1+S_2 \\ 1+S_1 & 1+S_2 & 1+S_3 \\ 1+S_2 & 1+S_3 & 1+S_4 \end{vmatrix} = \begin{vmatrix} 1+1+1 & 1+\alpha+\beta & 1+\alpha^2+\beta^2 \\ 1+\alpha+\beta & 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 \\ 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 & 1+\alpha^4+\beta^4 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \end{vmatrix}^2 = [(1-\alpha)(1-\beta)(\alpha-\beta)]^2$$

$D = (\alpha - \beta)^2 (\alpha + \beta - \alpha\beta - 1)^2$

$\because$ α & β are roots of the equation $ax^2 + bx + c = 0$

$$\Rightarrow \quad \alpha+\beta = \frac{-b}{a} \quad \& \quad \alpha\beta = \frac{c}{a} \quad \Rightarrow \quad |\alpha-\beta| = \frac{\sqrt{b^2-4ac}}{|a|}$$

$$D = \frac{(b^2-4ac)}{a^2}\left(\frac{a+b+c}{a}\right)^2 = \frac{(b^2-4ac)(a+b+c)^2}{a^4}$$ **Ans.**

Do yourself - 5 :

1. If the determinant $D=\begin{vmatrix}1 & 1 & 1\\ \alpha+\beta & \alpha^2+\beta^2 & 2\alpha\beta\\ \alpha+\beta & 2\alpha\beta & \alpha^2+\beta^2\end{vmatrix}$ and $D_1=\begin{vmatrix}1 & 0 & 0\\ 0 & \alpha & \beta\\ 0 & \beta & \alpha\end{vmatrix}$, then find the determinant D_2 such that $D_2=\frac{D}{D_1}$.

2. If $D_1=\begin{vmatrix}ab^2-ac^2 & bc^2-a^2b & a^2c-b^2c\\ ac-ab & ab-bc & bc-ac\\ c-b & a-c & b-a\end{vmatrix}$ & $D_2=\begin{vmatrix}1 & 1 & 1\\ a & b & c\\ bc & ac & ab\end{vmatrix}$, then D_1D_2 is equal to -

(A) 0 (B) D_1^2 (C) D_2^2 (D) D_2^3

3. Factorize : $\begin{vmatrix}3 & a+b+c & a^3+b^3+c^3\\ a+b+c & a^2+b^2+c^2 & a^4+b^4+c^4\\ a^2+b^2+c^2 & a^3+b^3+c^3 & a^5+b^5+c^5\end{vmatrix}$

(A) $[(a-b)^2(b-c)^2(c-a)^2(a+b+c)]$ (B) $[(a-b)(b-c)(c-a)(a+b+c)]$
(C) $[(a-b)^2(b-c)^2(c-a)^2]$ (D) None of these

4. If $\begin{vmatrix}1 & x & x^2\\ x & x^2 & 1\\ x^2 & 1 & x\end{vmatrix}=3$, then find the value of $\begin{vmatrix}x^3-1 & 0 & x-x^4\\ 0 & x-x^4 & x^3-1\\ x-x^4 & x^3-1 & 0\end{vmatrix}$

(A) 5 (B) 8 (C) 9 (D) 3

5. $\begin{vmatrix}a^2+b^2+c^2 & bc+ca+ab & bc+ca+ab\\ bc+ca+ab & a^2+b^2+c^2 & bc+ca+ab\\ bc+ca+ab & bc+ca+ab & a^2+b^2+c^2\end{vmatrix}$ is always

(A) non-negative (B) 0 (C) negative (D) Can't say

7. SPECIAL DETERMINANTS :

(a) Cyclic Determinant :

The elements of the rows (or columns) are in cyclic arrangement.

$$\begin{vmatrix}a & b & c\\ b & c & a\\ c & a & b\end{vmatrix}=-(a^3+b^3+c^3-3abc)=-(a+b+c)(a^2+b^2+c^2-ab-bc-ac)$$

$$=-\frac{1}{2}(a+b+c)\times\{(a-b)^2+(b-c)^2+(c-a)^2\}$$

$=-(a+b+c)(a+b\omega+c\omega^2)(a+b\omega^2+c\omega)$, where ω,ω^2 are cube roots of unity

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(b) Other Important Determinants :

(i) $\begin{vmatrix} 0 & b & -c \\ -b & 0 & a \\ c & -a & 0 \end{vmatrix} = 0$

(ii) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ac & ab \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$

(iii) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$

(iv) $\begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$

(v) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^4 & b^4 & c^4 \end{vmatrix} = (a-b)(b-c)(c-a)(a^2+b^2+c^2-ab-bc-ca)$

Illustration 9 : Prove that $\begin{vmatrix} 1 & \alpha & \alpha^2 \\ \alpha & \alpha^2 & 1 \\ \alpha^2 & 1 & \alpha \end{vmatrix} = -(1-\alpha^3)^2$.

Solution : This is a cyclic determinant.

$$\Rightarrow \begin{vmatrix} 1 & \alpha & \alpha^2 \\ \alpha & \alpha^2 & 1 \\ \alpha^2 & 1 & \alpha \end{vmatrix} = -(1+\alpha+\alpha^2)(1+\alpha^2+\alpha^4-\alpha-\alpha^2-\alpha^3)$$

$= -(1+\alpha+\alpha^2)(-\alpha+1-\alpha^3+\alpha^4)$

$= -(1+\alpha+\alpha^2)(1-\alpha)^2(1+\alpha+\alpha^2)$

$= -(1-\alpha)^2(1+\alpha+\alpha^2)^2 = -(1-\alpha^3)^2$

Do yourself - 6 :

1. The value of the determinant $\begin{vmatrix} ka & k^2+a^2 & 1 \\ kb & k^2+b^2 & 1 \\ kc & k^2+c^2 & 1 \end{vmatrix}$ is

(A) $k(a+b)(b+c)(c+a)$ (B) $kabc(a^2+b^2+c^2)$
(C) $k(a-b)(b-c)(c-a)$ (D) $k(a+b-c)(b+c-a)(c+a-b)$

2. Find the value of the determinant $\begin{vmatrix} a^2+b^2 & a^2-c^2 & a^2-c^2 \\ -a^2 & 0 & c^2-a^2 \\ b^2 & -c^2 & b^2 \end{vmatrix}$.

3. Prove that $\begin{vmatrix} a & b & c \\ bc & ca & ab \\ b+c & c+a & a+b \end{vmatrix} = (a+b+c)(a-b)(b-c)(c-a)$.

4. Let a, b, c be such that $b(a+c) \neq 0$. If $\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n+1}b & (-1)^n c \end{vmatrix} = 0$, then the value of n is :-

(A) Any odd integer (B) Any integer
(C) Zero (D) Any even integer

5. The product of all the values of α for which equations $(a-\alpha)x + by + c = 0$, $cx + (a-\alpha)y + b = 0$ and $bx + cy + a - \alpha = 0$ are consistent, when $a,b,c > 0$, is -

(A) zero (B) positive (C) negative (D) one

8. CRAMER'S RULE (SYSTEM OF LINEAR EQUATIONS) :

Simultaneous linear equations

Consistent
(at least one solution)

Inconsistent
(no solution)

Exactly one solution
or
Unique solution

Trivial solution
i.e. All variable
zero is the
only solution

Infinite solutions

Non trivial solution

At least one
non zero variable
satisfies the system

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(a) Equations involving two variables :

(i) Consistent Equations : Definite & unique solution (Intersecting lines)

(ii) Inconsistent Equations : No solution (Parallel lines)

(iii) Dependent Equations : Infinite solutions (Identical lines)

Let, $a_1x + b_1y + c_1 = 0$

$a_2x + b_2y + c_2 = 0$ then :

(1) $\dfrac{a_1}{a_2} \neq \dfrac{b_1}{b_2} \Rightarrow$ Given equations are consistent with unique solution

(2) $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} \neq \dfrac{c_1}{c_2} \Rightarrow$ Given equations are inconsistent

(3) $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \dfrac{c_1}{c_2} \Rightarrow$ Given equations are consistent with infinite solutions

(b) Equations Involving Three variables :

Let $a_1x + b_1y + c_1z = d_1$ .... (i)

$a_2x + b_2y + c_2z = d_2$ .... (ii)

$a_3x + b_3y + c_3z = d_3$ .... (iii)

Then, $x = \dfrac{D_1}{D},\ y = \dfrac{D_2}{D},\ z = \dfrac{D_3}{D}$.

Where $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$; $D_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$; $D_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$ & $D_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$

Note :

(i) If $D \neq 0$ and atleast one of $D_1, D_2, D_3 \neq 0$, then the given system of equations is consistent and has unique non trivial solution.

(ii) If $D \neq 0$ & $D_1 = D_2 = D_3 = 0$, then the given system of equations is consistent and has trivial solution only.

(iii) If $D = 0$ but atleast one of D_1, D_2, D_3 is not zero then the equations are inconsistent and have no solution.

(iv) If $D = D_1 = D_2 = D_3 = 0$, then the given system of equations may have infinite or no solution.

Note that In case $\left.\begin{matrix} a_1x + b_1y + c_1z = d_1 \\ a_1x + b_1y + c_1z = d_2 \\ a_1x + b_1y + c_1z = d_3 \end{matrix}\right\}$ (Atleast two of d_1, d_2 & d_3 are not equal)

$D = D_1 = D_2 = D_3 = 0$. But these three equations represent three parallel planes. Hence the system is inconsistent.

(c) Homogeneous system of linear equations :

If x, y, z are not all zero, the condition for

$a_1x + b_1y + c_1z = 0$
$a_2x + b_2y + c_2z = 0$
$a_3x + b_3y + c_3z = 0$

to be consistent in x, y, z is that $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$

Remember that if a given system of linear equations have **Only Zero** Solution for all its variables then the given equations are said to have **TRIVIAL SOLUTION**.

Illustration 10 : Find the nature of solution for the given system of equations :
$x + 2y + 3z = 1;\ 2x + 3y + 4z = 3;\ 3x + 4y + 5z = 0$

Solution :

$$D = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} = 0$$

Now, $D_1 = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 3 & 4 \\ 0 & 4 & 5 \end{vmatrix} = 5$

$\because$ $D = 0$ but $D_1 \neq 0$

Hence no solution. **Ans.**

Illustration 11 : Find the value of λ, if the following equations are consistent :
$x + y - 3 = 0;\ (1 + \lambda)x + (2 + \lambda)y - 8 = 0;\ x - (1 + \lambda)y + (2 + \lambda) = 0$

Solution : The given equations in two unknowns are consistent, then $\Delta = 0$

i.e. $\begin{vmatrix} 1 & 1 & -3 \\ 1+\lambda & 2+\lambda & -8 \\ 1 & -(1+\lambda) & 2+\lambda \end{vmatrix} = 0$

Applying $C_2 \to C_2 - C_1$ and $C_3 \to C_3 + 3C_1$

$\therefore$ $\begin{vmatrix} 1 & 0 & 0 \\ 1+\lambda & 1 & 3\lambda-5 \\ 1 & -2-\lambda & 5+\lambda \end{vmatrix} = 0$

$\Rightarrow$ $(5+\lambda) - (3\lambda-5)(-2-\lambda) = 0$ $\Rightarrow$ $3\lambda^2 + 2\lambda - 5 = 0$

$\therefore$ $\lambda = 1, -5/3$

Illustration 12 : If the system of equations $x + \lambda y + 1 = 0$, $\lambda x + y + 1 = 0$ & $x + y + \lambda = 0$. is consistent, then find the value of λ.

Solution : For consistency of the given system of equations

$$D = \begin{vmatrix} 1 & \lambda & 1 \\ \lambda & 1 & 1 \\ 1 & 1 & \lambda \end{vmatrix} = 0$$

$\Rightarrow$ $3\lambda = 1 + 1 + \lambda^3$ or $\lambda^3 - 3\lambda + 2 = 0$

$\Rightarrow$ $(\lambda-1)^2(\lambda + 2) = 0 \Rightarrow \lambda = 1$ or $\lambda = -2$ **Ans.**

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Do yourself -7 :

1. Find nature of solution for given system of equations

$2x + y + z = 3$; $x + 2y + z = 4$; $3x + z = 2$

2. If the system of equations $x + y + z = 2$, $2x + y - z = 3$ & $3x + 2y + kz = 4$ has a unique solution, then

(A) $k \neq 0$ (B) $-1 < k < 1$ (C) $-2 < k < 1$ (D) $k = 0$

3. The system of equations $\lambda x + y + z = 0$, $-x + \lambda y + z = 0$ & $-x - y + \lambda z = 0$ has a non-trivial solution, then possible values of λ are -

(A) 0 (B) 1 (C) -3 (D) $\sqrt{3}$

4. Consider the system of linear equations : $x_1 + 2x_2 + x_3 = 3$, $2x_1 + 3x_2 + x_3 = 3$, $3x_1 + 5x_2 + 2x_3 = 1$ The system has

(A) Infinite number of solutions (B) Exactly 3 solutions

(C) A unique solution (D) No solution

5. The number of values of k for which the linear equations $4x + ky + 2z = 0$, $kx + 4y + z = 0$, $2x + 2y + z = 0$ possess a non-zero solution is :

(A) 1 (B) zero (C) 3 (D) 2

6. If the trivial solution is the only solution of the system of equations $x - ky + z = 0$, $kx + 3y - kz = 0$, $3x + y - z = 0$ Then the set of all values of k is :

(A) $\{2, -3\}$ (B) $R - \{2, -3\}$ (C) $R - \{2\}$ (D) $R - \{-3\}$

7. Solve the following system of equations $x + y + z = 5$, $2x + 2y + 2z = 7$, $3x + 3y + 3z = 6$

(A) Unique solution (B) infinite solution (C) No solution (D) None of these

8. The number of values of k for which the system of equations

$(k + 1)x + 8y = 4k$

$kx + (k + 3)y = 3k - 1$

has infinitely many solutions is

(A) 0 (B) 1 (C) 2 (D) Infinite

9. The values of 'λ' for which the system of equations

$(1 - \lambda)x + 3y - 4z = 0$

$x - (3 + \lambda)y + 5z = 0$

$3x + y - \lambda z = 0$

possesses non-trivial solution, is/are

(A) -1 (B) 0 (C) 1 (D) 2

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